

KAVACH AI

India's AI-Powered Digital Safety Shield

Defeating Counterfeiting, Fraud & Digital Arrest Scams

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Track: Smart Cities / Public Safety / Digital Trust / Geospatial Law Enforcement
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1. Problem Statement & Executive Context

The Growing Public Safety Crisis:

India registered over **1.14 million cybercrime complaints** in 2023 — a 60% surge from 2022. Among the most dangerous emerging threats are '**Digital Arrest**' scams, where victims are psychologically held hostage over video calls by criminals impersonating CBI, ED, or Police officers. The Ministry of Home Affairs reported citizens defrauded of over **Rs 1,776 crore** in the first nine months of 2024 alone.

Concurrently, counterfeit currency (Fake Indian Currency Notes — FICN) continues to undermine financial systems, with high-quality Rs 500 fake notes bypassing traditional manual detection. Existing law enforcement frameworks are reactive, stepping in only post-complaint rather than neutralizing threats at the point of contact.

2. Proposed Solution — KAVACH AI

KAVACH AI is an integrated, multi-agent AI public safety platform designed to equip citizens, financial institutions, and law enforcement agencies (LEO) with real-time proactive tools to detect, disrupt, and neutralize digital fraud and counterfeit currency.

- **Citizen Fraud Shield:** Multilingual conversational advisory in 12 regional languages providing real-time call/message threat verdicts & 1-click reporting to NCRB 1930.
- **Digital Arrest Scam Detector:** ML-powered transcript & speech classifier trained on 15 call categories to detect impersonation, fear tactics, and urgency indicators.
- **Counterfeit Currency Agent:** Computer vision forensics analyzing microprint, intaglio edge sharpness, HSV color profiles, and printer frequency artifacts across denominations.
- **Fraud Network Graph AI:** Graph algorithms (Louvain Community Detection, PageRank) mapping money mule clusters, suspect nodes, and organized campaign infrastructure.
- **Geospatial Crime Hotspots:** Spatial analytics (DBSCAN & Kernel Density Estimation) identifying cybercrime hotspots and guiding police patrol prioritization.
- **LEO Command & Court Evidence:** End-to-end case management generating court-admissible PDF intelligence packages.

3. Hero Feature Showcase — The 3 Core Dashboards

KAVACH AI features **three distinct, role-tailored hero dashboards** that clearly demarcate citizen protection, police investigations, and system administration.

Dashboard Role	Target Users & Core Goal	Exclusive Hero Features
CITIZEN DASHBOARD Public Safety Shield	Citizens, families, and financial customers looking to verify suspicious contacts or notes.	• Scam Text & Call Transcript Scanner • Currency Note Computer Vision Forensics • Media Forensics (Deepfake & Voice Cloning) • 12 Regional Language RAG Chatbot • 1-Click NCRB 1930 Cybercrime Reporting
LEO DASHBOARD Command & Investigation	Law Enforcement Officers, Cyber Cells, & Financial Crime Investigators.	• Real-Time Case Management & Status Vault • NetworkX Fraud Ring Graph Analytics • DBSCAN / KDE Geospatial Crime Hotspot Map • Suspect & Mule Account Profiler • Court-Admissible PDF Evidence Exporter
ADMIN DASHBOARD System Governance	System Administrators, MHA Authorities, & Network Operators.	• User Directory & Role Control (Citizen→LEO) • Emergency SMS & In-App Broadcast Center • Real-Time System Audit Log Inspector • FastAPI Runtime Health Diagnostics • Dynamic System Settings & Threat Intel

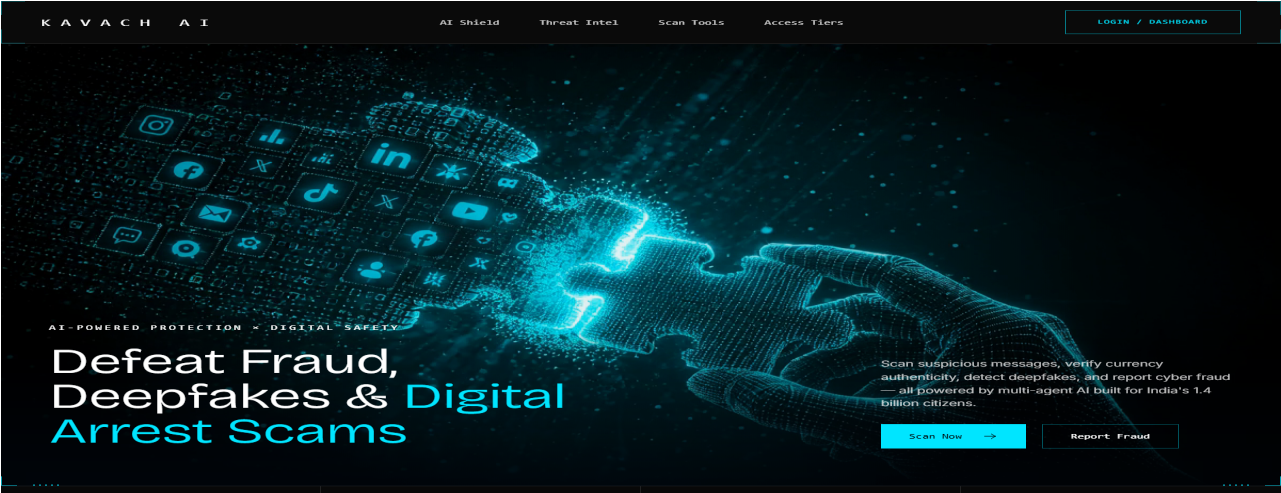
Role-Based Security & Isolation Matrix:

Access control is strictly enforced at both the FastAPI backend middleware (``jwt_bearer`, `get_current_active_user``) and frontend DOM views (``.LEO-only``, ``.ADMIN-only``). Citizens cannot view police case tables, while LEOs have investigation tools, and Admins maintain full platform oversight.

4. Live Dashboard Visual Tour

Actual screenshots captured directly from the live local execution environment:

1. Citizen Safety Shield & Scan Portal View:



2. Law Enforcement Officer (LEO) Command View:

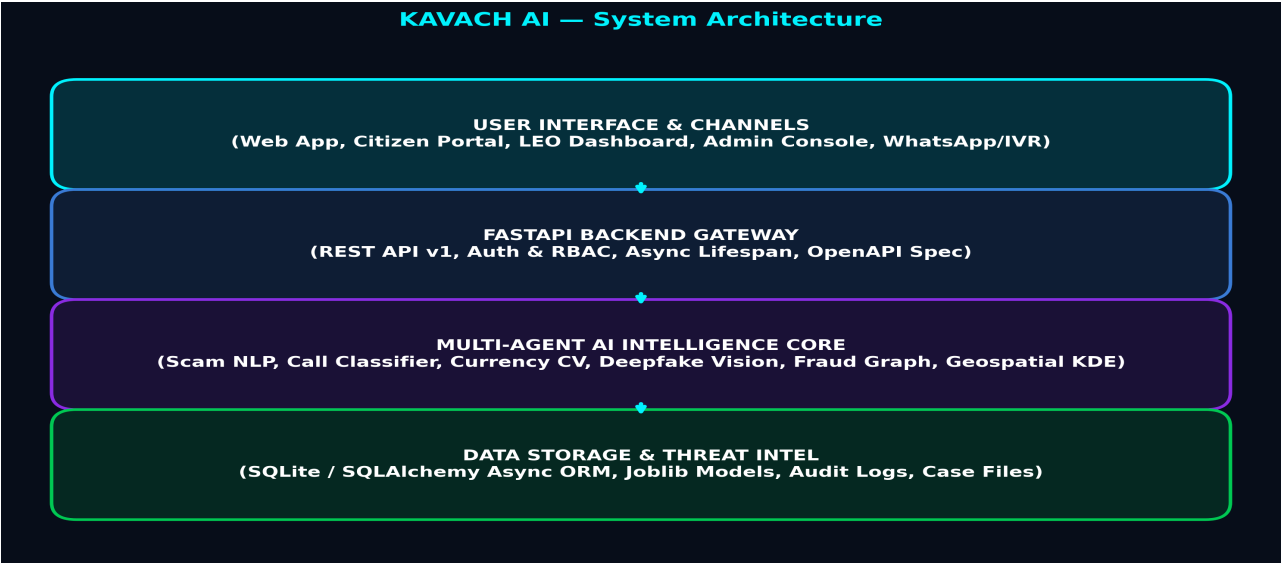


3. Admin Platform Governance & Control View:

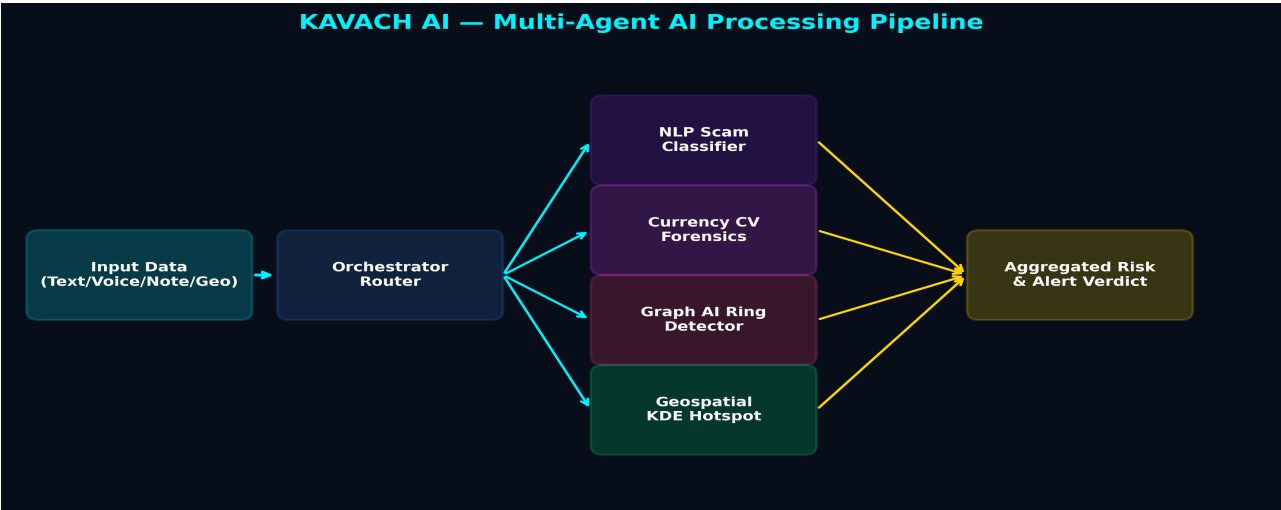


5. System Architecture & Processing Workflows

System Architecture Diagram:



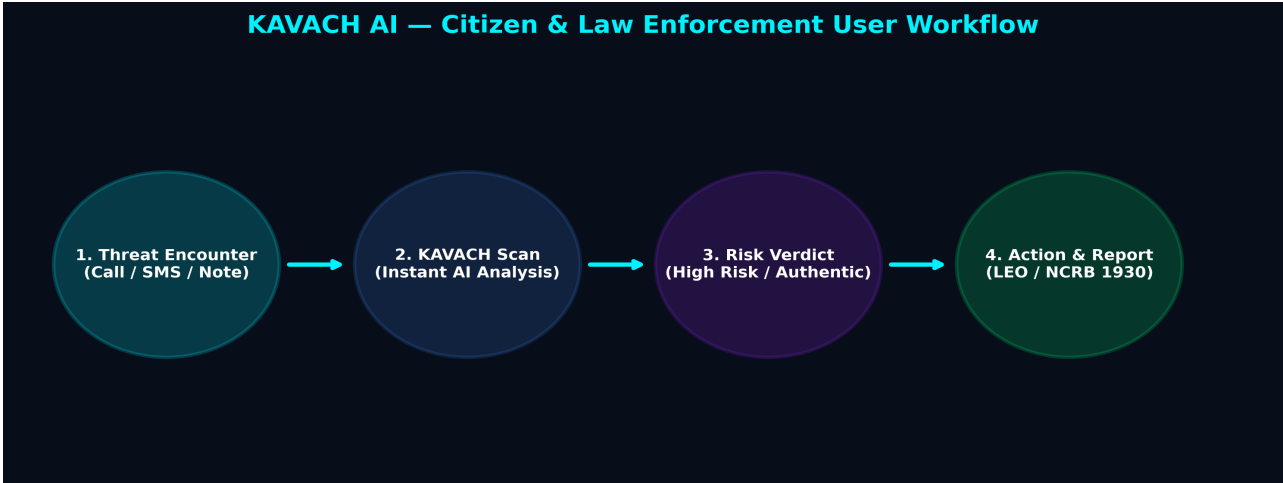
Multi-Agent AI Processing Pipeline:



6. Technology Stack & Technical Matrix

Component	Technologies Used	Key Purpose
Frontend	HTML5, Vanilla JS, TailwindCSS, Iconify, GSAP	Responsive Citizen, LEO & Admin Dashboards
Backend API	FastAPI, Uvicorn, Python 3.11, Pydantic v2	Async REST APIs & Microservices Layer
AI & ML Models	Scikit-Learn, NetworkX, SciPy, NumPy, Joblib, Whisper	Scam NLP, Call ML, Graph AI, Geospatial
Computer Vision	OpenCV, Pillow (PIL), FFT Frequency Analysis	Currency Note Forensics & Deepfake Detection
Database & Auth	SQLite, SQLAlchemy Async ORM, PyJWT, Bcrypt	Persistent Data & Role-Based Access

User Processing Journey Workflow:



7. Future Scope & Strategic Roadmap

- **WebRTC Live Call Interception:** Integrating real-time audio stream monitoring directly on mobile devices to warn citizens mid-call during active digital arrest attempts.
- **Direct Carrier & WhatsApp Cloud API Integration:** Webhook bridges connecting telecom service providers and Meta WhatsApp Business API for instant threat blocking.
- **Distributed Graph & Database Scaling:** Upgrading SQLite persistence to enterprise PostgreSQL + Neo4j for multi-billion transaction graph analysis.
- **Hardware UV Sensor Integration:** Calibrating computer vision models for specialized UV/IR camera hardware on banking note counting machines.

8. Socio-Economic Impact & Evaluation Alignment

KAVACH AI addresses the core evaluation criteria of the ET AI Hackathon 2026:

- **Innovation (25%):** Multi-agent fusion uniting NLP, Computer Vision, Graph AI, and Geospatial analytics into a single public safety shield.
- **Business & Social Impact (25%):** Safeguarding citizen life savings against psychological digital arrests and preventing counterfeit currency proliferation.
- **Technical Excellence (20%):** Self-contained, zero-external-dependency production architecture with 25+ REST endpoints and trained ML models.
- **Scalability & UX (30%):** Async FastAPI backend, 12 regional languages, and intuitive dark-mode glassmorphism interface.

Conclusion:

Project **KAVACH AI** by Team **BitBuddies** provides a production-grade, multi-agent AI safety shield that shifts public safety from reactive post-crime reports to real-time predictive threat neutralization for citizens, law enforcement, and financial institutions.